With the brake pipe broken back of the brake valve branch, plug the pipe toward the brake valve and run the brake pipe pressure through the engine signal pipe to the train by connecting the brake pipe and signal hose at the pilot and at the rear of the engine. If the engine has no signal system, as in freight service, it is possible to charge the train pipe through the brake cylinder pipes by cutting out the engine brake cylinders and connecting the brake cylinder pipe at the rear of the engine with the brake pipe at the front of the tender. In this manner the brake valve can be allowed to remain on lap position and by moving the independent brake valve to application position the distributing valve will admit main reservoir pressure toward the brake cylinders, but instead of entering the cylinders it will flow to the brake pipe through the connection made between the engine and tender. The safety valve of the distributing valve can then be screwed down and the reducing valve can be adjusted to maintain any pressure desired in the brake pipe. This will be understood to be merely an emergency brake and the distributing valve must perform the work of a feed valve which it is not designed to do; it may, however, be sufficiently sensitive to prevent leakage from applying the brake, and to apply the brake the independet valve must be carefully moved to release position to exhaust brake pipe pressure for an application of the train brakes, without resulting in an emergency application.

This method of using the brake cylinder pipes and the distributing valve to make a connection to the brake pipe on the cars also applies to a case where the brake valve branch of the brake pipe may be broken off or where the reservoir pipe may be broken off at the brake valve when there is a feed valve attached to the H 6 brake valve or in any case where the automatic brake valve could not be used at all. Some air brake men suggest the use of the dead engine feature for charging the brake pipe from the main reservoir in case of a disabled automatic brake valve by removing the check valve of the dead engine feature which permits air to flow through the choke fitting to supply the brake pipe, while we question the practicability of such a [illegible] trains short enough so that it can be charged through this opening, no emergency brake will be had and may be applied by opening the [illegible] cut out cock, assuming that the brake pipe branch is broken off [illegible].

Under ordinary circumstances a broken reservoir pipe at the brake valve can be temporarily repaired by plugging both ends of the broken pipe and using running position of the brake valve for a train brake release.

Another method of charging a train with a broken off brake pipe branch to the brake valve is to allow the automatic brake valve to remain in lap position, cut out the distributing valve with the stop cock in the supply pipe and remove the equalizing valve of the distributing valve and charge the train with the independent brake valve in quick application position. The brake to be applied by moving the independent valve to release position, or in emergencies by opening the brake valve cut out cock. With the return spring casing screw removed from the independent brake valve body, the valve handle will remain in quick application position, but as this involves the taking apart of some of the apparatus we do not exactly recommend it, but mention it merely as air brake information.

If the brake pipe is broken or burst under the tender, the signal pipe under the tender must be used to make a brake pipe connection by crossing the hose front and back of the tender. The signal whistle will be inoperative, but this is generally recognized as the correct method to be employed on a tender or car on a passenger train in the event of a broken brake pipe.

If the distributing valve supply pipe is broken off at the distributing valve, close the supply pipe stop cock and proceed to the terminal without an engine brake. If broken off between the reservoir pipe and the cut out cock, close the main reservoir cut out cock to maintain main reservoir pressure while the pipe is being plugged and then proceed without a driver brake.

With a broken distributing valve supply pipe on a lone engine, several methods for obtaining a brake on the engine have come to our notice; one is, if the distributing valve has a quick action cylindre cap is to remove the emergency side valve and the brake cylinder check valve and carry the brake valve on lap or in service position with the brake pipe drained, then when an application is desired, placing the automatic brake valve in release position will admit brake pipe pressure to the brake cylinders and apply the brake. The release to be made by moving the brake valve to emergency position. If the distributing valve has a plain cylinder cap, the cylinder cap gasket can be removed and a slot cut through the gasket between the brake pipe and brake cylinder ports and the brake applied and released in the same manner as when the emergency slide valve and check valve have been removed from the quick action cylinder cap.

In the foregoing it will be understood that the broken supply pipe is separated in such a manner that the two broken ends can not be removed and a connection made with two signal hose, for in case this can be done there will be no necessity for proceeding without an engine brake.

If the brake cylinder pipe is broken off at the distributing valve, move the independent brake valve to release position to prevent a waste of main reservoir pressure and at the first opportunity close the stop cock in the distributing valve supply pipe until repairs can be made at a shop.

If the brake cylinder pipe is broken under the tender, close the stop cock in the brake cylinder pipe at the rear of the engine and only the tender brake will be affected.

If the brake cylinder hose between the engine and tender are bursted, close the same stop cock until such a time as the hose can be removed without causing a delay.

In case of a broken brake pipe branch to the distributing valve, lap the automatic brake valve and plug the broken pipe toward the main brake pipe to prevent a brake pipe leak, then release brake on the engine with the independent brake valve and thereafter use the independent valve to operate the engine brake while the automatic brake valve is used to operate the train brakes.

In the event of a broken application cylinder pipe, plug the connection at the distributing valve and proceed. There will be no independent brake, but the automatic brake will not be affected otherwise that the engine brake can not then be released independently of the train brakes.

In the case of a broken distributing valve release pipe no repairs need be made, as only the holding feature of the automatic brake will be lost. The independent brake can also be applied in quick application position, but will release as soon as the valve handle is returned to lap or slow application position.

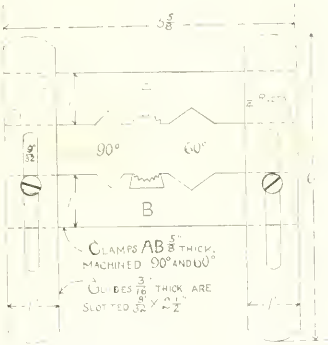
Should this occur in yard shifting service where the use of the independent brake is preferable to the automatic, the release pipe connection at the distributing valve can be plugged and the return spring casing screw removed from the independent brake valve body and the independent valve carried in release position until the correct repairs can be made.

We believe that the foregoing covers the subject fairly well and that a study of this matter will enable a man to bring a train to a terminal under ordinary conditions of broken air pipes on the locomotive.

Improved Vise Clamps

By WILLIAM DURANT, Somerville, Mass.

The sketch herewith shows a handy device for holding small vise work. No extended description is necessary. Simply lay the guides of the clamp on the jaws of a vise then tighten the work between the jaws of the clamp. For injector repairs, or work such as holding a pipe, the toothed jaws shown are useful. These jaws may be cut in the clamps, shown at A, or they may be made of tempered steel, inserted, shown in clamp B. The two clamps are practically duplicate, and can be made of machine steel, case-hardened. The slotted guides are secured by rivets, flush with the top sides of the clamps which are planed, also by one fourth machine screws, allowing quick opening and closing. For the bench man who has pride in saving corners of square and hexagonal brass work, the device will be found of much value.


5 5/8
1/4 Rivets
90°
60°
9/32
B
Clamps AB 5/8" thick, machined 90° and 60°
Guides 3/16" thick are slotted 9/32" × 2 1/2"
1"
1"

DETAILS OF IMPROVED VISE CLAMPS.

The Defects in Shoes and Wedges

Improper Attention the Real Cause

By C. PENDLETON, Foreman of Engines, Washington & Old Dominion Railway

I have read with interest the article in the May issue of RAILWAY AND LOCOMOTIVE ENGINEERING, by Mr. Scheetz, of Shamokin, Pa., having reference to defects in shoes and wedges. I differ from him, in that the different supposed defects mentioned in his article are defects in the shoes and wedges, and are not defects in the shoes and wedges at all, but are due to improper attention being given to the running or shop repairs of the engines.

Where he attributes the varying hardness of the castings as being directly the cause of not being able to keep the drivers square and in tram, by causing uneven wear, it is easier to obtain a clear, clean casting of cast iron than it is of steel, at a cheaper cost owing to the fact that enough allowance is made in the casting to permit of a face of good metal being secured, and the cost of the waste is very small compared with what that of steel would be. Also, the cost of machining the cast iron is very small compared with that of machining steel.

It has been my understanding for several years that the reason for using cast iron for shoes and wedges is to replace the wear, that does take place naturally, and also the unnatural wear, at as small cost as possible. The first cost being small as compared with steel and on account of being softer than the material of the frames and the driving boxes the shoes and wedges take up or use up about 97 per cent of the wear, natural and otherwise. Thus the wear can be replaced in this manner cheaply where if steel shoes and wedges were used it would mean a worn box in shoe and wedge way equal to the wear on shoes and wedges, which can be readily seen is not at all desirable. The wear also differs on the different wheels and is not due so much, I think, to defects in the castings of the shoes and wedges as to improper attention being given to the writing up of work reports by the different engineers, especially in pool service the wedge work done by the engineers, and the work done to correct defects as shown in the work reports by the mechanics in the roundhouses at terminal or division points.

As to the fact of the bottom part of the wedge pulling out, this is very seldom if ever the case under wedge design of the past ten years, and if it does happen is due to improper design of the wedge bolt position, in so far as the hole is concerned than to defective casting. In all cases coming under my observation, which are very numerous, it has always been the case that the head of the wedge bolt was either pulled or broken off.

The breaking of the flanges on the shoes and wedges is not due to defective castings so much, I think, if at all, as to badly worn pedestal jaws and engines having excessive lateral in the driving wheels, usually excessive lateral.

The tremendous blows struck when the engine is passing over what is called the center instead of showing defective compressibility of the casting by it being battered out of shape, shows the effect of improper attention being paid to the engines in adjusting the shoes and wedges to take up natural wear. The results of which could be entirely eliminated by proper adjustment of the wedges, which should have an inspection every 15 days, whether reported as being in bad condition by the engineer or not, keeping in mind the fact that the brunt of the natural wear of the parts of the engine falls on the shoes and wedges, as well as the rod brasses.

Where Mr. Scheetz contends that the friction in the shoes and wedges is not purely sliding friction, but pounding, he would leave the impressing that it is pounding alone. I think the readers will agree with me here, that if proper means are used to take up the sliding friction, which should be the only real friction in the shoes and wedges, that there would be no pounding present. If the shoes and wedges are kept adjusted properly there should be no movement to cause excessive friction other than the movement of the boxes up and down, which is sliding friction. The lost motion arising from poor adjustment causes the pounding and should not be credited to the casting as being defective.

In so far as the use of brass for shoes and wedges goes the cost is absolutely prohibitive, although good results have been obtained in some cases by using brass filling on the driving boxes to work against the cast iron shoes and wedges. This has not met with general approval.

In case the engines are permitted to run in such condition to batter out of shape the shoes and wedges, it would still be better to use cast iron than steel, for if steel was used it would spread out more than the cast iron and the burrs resulting would cause excessive wear on the shoe and wedge way sides of the driving boxes, and in most cases on long fast runs would result in what is termed stuck wedges.

The wear caused in the shoe and wedge way of the box in the wearing off or cutting down the burrs on the shoes and wedges, if of steel, would in most cases mean a new pattern for both shoes and wedges when the engine goes in for a general overhauling as the shoe and wedge way in the box would have to be planed out laterally, as well as on the face to obtain good bearing surface, for the shoes and wedges which would necessitate a wider pattern of both shoe and wedge. This wear on the box is not present to sufficient extent to be very noticeable, or to be paid attention to in usual run of locomotive repair shops, where the cast iron shoe and wedge is used, as the wear comes on the shoe and wedge practically altogether. When the cast iron shoe and wedge is used it is a very unusual thing to plane out the shoe and wedge way of the driving boxes laterally if it is ever done before the driving boxes are scrapped.

Catechism of Railroad Operation

Locating Pounds and Keying Up Rods, and Disconnecting for Broken Rods and Levers

NEW SERIES.

Third Year's Examination.

(*Continued from page 123, April, 1915.*)

Q. 186.—How do you place engine to locate pounds in main rod brasses and why in that position?

A.—Place engine on bottom quarter on side to be tested, set brake, admit steam to cylinder and work reverse lever back and forth, watch brasses. We place the engine on bottom quarter so as to have the pin between two rigid points, and then any lost motion in brasses will show before box would move or the wheel slip.

Q. 187.—Where do you place engine to key *back end* of main rod and why there?

A.—On dead-center, so as to key brasses against largest part of crank pin.

Q. 188.—Where do you place engine to key front end of main rod brasses and why there?

A.—On top or bottom quarter. Generally *bottom quarter* so as to key against largest part of pin, and on bottom quarter because it is easier to get at set screw that holds front end key.

Q. 189. What would you do if the main rod broke?

A. Remove broken parts and disconnect valve rod and block valve to cover admission ports, block crosshead, remove cylinder cocks, put collar on main pin and provide for lubrication.

Q. 190.—What would you do if the piston rod broke off at crosshead?

A.—Remove broken parts, disconnect valve rod, block valve to cover the admission ports and proceed. Note:—Most likely the piston would have gone out through the front cylinder head.

Q. 191. What would you do if front cylinder head broke?

A.—Disconnect valve rod, block valve with back admission port slightly open, remove back cylinder cock. Note:—Do not take down main rod. Note: If in dusty weather, would cover front end of cylinder with gunny sack or boards. Another way: Disconnect valve rod, block valve centrally on its seat, remove back cylinder cock, board up front end of cylinder, placing a swat of waste saturated with oil in cylinder. Another way: Disconnect valve rod, block valve centrally on its seat, disconnect main rod, put collar on main pin and block crosshead.

Q. 192. With the Walschaerts valve motion explain how you could disconnect for broken eccentric crank, eccentric rod, or link foot.

A.—Disconnect eccentric rod and remove other broken parts, disconnect back end of radius rod and block link block in center of the link. Note:—When blocking link block in link of Walschaerts valve gear block above and below it in slot of link.

Q. 193.—What would you do if link trunnion pin broke?

A.—Take down eccentric rod, disconnect back end of radius rod and block link block in center of link, then wedge the link in place by driving wedge between trunnion pin bracket and link on side where the trunion pin is broken.

Q. 194.—What would you do when link trunnion pin bracket breaks?

A.—Take down eccentric rod, disconnect both ends of radius rod and take link and radius rod out, block valve centrally on seat, provide for lubrication and free circulation of air in cylinder. Note:—If link cannot be readily removed, support forward end of radius rod and chain link so as it will carry, allowing back end of radius rod to carry with link block at bottom of link.

Q. 195.—What would you do with broken valve stem or valve yoke of Walschaerts valve motion?

A.—Block valve centrally on seat, disconnect forward end of radius rod and support it underneath the running board, provide for lubrication and free circulation and proceed. Note:—With link block in center of link and combination lever (lap and lead lever) at right angles to it the valve will be centrally on its seat. Note:—To secure the valve centrally where yoke or stem is broken, locate the central position, clamp valve stem then remove the vacuum valve and block it against the valve, or if it is a piston valve, remove the front head to valve chamber and cut piece of board (that is nearly as wide as diameter of valve chamber) to fit between valve in central position and head of valve chamber. Note:—Some desire as an extra precaution to stop all motion of radius rod when front end of it is disconnected and supported up. To do this you may remove eccentric rod, or disconnect back end of radius rod and block link block in center of link. Note:—When the forward end of radius rod is supported and eccentric rod is taken down, the link block does not always move freely in link and at times bothers in handling the reverse lever, consequently the better way to stop motion of radius rod is to disconnect back end of it and block link block in center of link.

Q. 196.—What would you do in case of broken reversing lever? Reach rod or reversing arm, or lifting shaft, Walschaerts valve gear?

A.—If reversing arm extended up through running board and was not broken, could block forward and back of it in slot in running board to hold link blocks at the desired point of cut off, otherwise would have to block above and below both link blocks in slot of link to hold link blocks at the desired point of cut off, necessary to start and handle train.

Q. 197.—What would you do in case of a broken radius arm or radius arm hanger?

A.—Remove the broken parts and block link block in slot of link to get the desired point of cut off to handle train. Note:—With the Walschaerts valve gear. Where one link block is blocked as above the other side may be worked at a longer or shorter cut off as desired.

Q. 198.—What would you do in case of a broken radius rod?

A.—Remove the broken part from the combination lever—support the forward end of radius rod, secure valve centrally on seat, provide for lubrication and free circulation in cylinders and proceed. Note:—Absolute safety is provided by disconnecting back end of radius rod and blocking link block in center of link.

Q. 199.—What would you do in case of a broken lap and lead lever, union link or crosshead arm?

A.—Remove broken parts necessary, block valve centrally on its seat, support front end of radius rod, provide for lubrication and free circulation and proceed. Note:—Where lap and lead lever is not broken the lower end of it may be secured to back cylinder cock. Note:—Many engines have the front end of radius rod constructed so it can be connected to back end of valve rod, when the lap and lead lever is taken down. On such engines it is a good plan in accidents of this kind to make the connection between radius rod and valve rod. This will give enough movement to valve to provide the lubrication to cylinder but will not aid much in handling the train because the eccentric only moves the valve far enough to open ports wide if the valve had no lap, and the engine working full stroke, and as the cut off is shortened by hooking up of lever so is travel of valve shortened and with lever hooked up the lap of valve would prevent much port opening and the angularity of main rod would cause an unequal distribution, so the actions of steam on disabled side would be erratic at the best, and the ports would only be open a very little and for only an instant when cranks were passing the quarters on the disabled side.

Catechism of Railroad Operation

Broken Frames, Crank Pins, Rods, Equalizers and Wheel Tires

Third Year's Examination.

(Continued from page 267, August, 1915.)

Q. 234.—Explain how you would handle engine with broken frame rails, ahead or back of main driver.

A.—The old rule is "with bottom frame rail broken at any place or top frame rail broken ahead of main box, do not try to handle any train unless you disconnect on the disabled side, and with top frame rail broken back of main driver handle train but do not work as second engine in double heading?"

Q. 235.—In case of broken side rods, what should be done? Why?

A.—Remove disabled rod or rods and the corresponding rods on the other side, becaues it is necessary to have side rods on both sides of engine so one side will help the other pass the dead center. NOTE.—With side rods up on only one side of engine, at time that side is passing center if main wheel should start to slip (or any of the wheels should be higher or lower than the other ones) the rest of wheels are as liable to turn the other way as they are to keep revolving in proper direction and pins or rods must break.

Q. 236.—What would you do if piston rod gland or valve stem gland broke?

A.—Cut two pieces of sheet iron or of an old scoop shovel, with slots that would go over studs and rod, slide one in from each side and tighten against gland, with the studs.

Q. 237.—What would you do if the intermediate section of side rod or main pin broke on "Brooks" consolidation engine? (Long main rod.)

A.—Take down all rods both sides, clamp valves centrally on seats, block crossheads, remove cylinder cocks and be towed in. NOTE.—Eccentrics are on No. 2 axle and main rod connected at No. 3 wheels.

Q. 238.—What would you do if stud broke in valve stem or piston rod packing gland?

A.—Would brace the gland if possible, if not, go right along.

Q. 239.—If metallic packing gives out on road and you have none to replace it what would you do?

A.—Wrap a piece of bell cord with candle wicking and pack piston rod or valve stem with it.

Q. 240.—If forward section side rod or front driving pin broke on Mogul engine what would you do?

A.—Take down all side rods and go in with one pair of wheels.

Q. 241.—If back section side rod breaks on Mogul or Consolidation engine, what would you do? Or forward section on Consolidation?

A.—Take down broken parts and the corresponding section on other side.

Q. 242.—If middle section side rod breaks on Consolidation engine, what?

A.—Take down all side rods both sides and go in with one pair of wheels.

Q. 243.—If main crank pin broke, what would you do?

A.—Take down all side rods, both sides, and main rod on the disabled side (or carry it in guides). Secure valve centrally on seat, block crosshead and remove cylinder cocks. NOTE.—Some types of engines are so constructed that crosshead bars will not clear pin on forward driver (with crosshead blocked). With this type of engine you will have to sling the forward pair of drivers and carry them clear of rail. To do this run both wheels up on wedges, block between pedestal binder and oil cellar and on top of frame under spring saddle to relieve weight, get off the wedges and counter balances will turn wheels so that pins will be on top eighths. As an extra precaution you may chain around spokes and across above frame rail, but it is not necessary.

Q. 244.—How would you relieve the weight from the journal on engines having large cross spring resting on top of driving boxes under frame rail?

A.—The only way possible is to run wheel up on wedge and block under box on top of pedestal binder.

Q. 245.—Do all Mogul and ten-wheel engines have knuckle joint in side rod back of main pin?

A.—No, some types have the knuckle joint ahead of main pin.

Q. 246.—If forward tire was very loose or broken on a Mogul or Consolidation engine, how would you handle it?

A.—Run the disabled wheel on wedge to raise box as high as possible in the jaws, remove oil cellar and use two blocks between binder and journal (putting one on top of pedestal binder with grain of wood across binder and the one on top of it with groove cut for journal bearing on ends of grain of wood), furnish lubrication for journal with bar of soap, hard cup grease or oily waste, relieve weight from box by blocking between frame and spring saddle, cut out driver brake. NOTE.—If rods are [illegible] do not take them down.

Q. 247.—What if main driving tire [illegible] very loose or broken?

A.—Run disabled wheel up on wedge to raise box as high as possible [illegible] remove the oil cellar, block [illegible] binder under journal, relieve [illegible] by blocking on top of frame under spring saddle, and cut out driver brake. NOTE.—If arch equalizers are used, block under ends of arch equalizers, on top of bottom frame rail. If underhung springs, chain ends of equalizers up to the bottom frame rail. NOTE.—You may have to take down binder to get oil cellar out.

Q. 248.—With intermediate driving tire very loose or broken, what would you do?

A.—Run disabled wheel up on wedge, remove oil cellar, block between binder and journal, relieve weight from box and cut out driver brake.

Q. 249.—With back driving tire very loose or broken?

A.—Run disabled wheel up on wedge to raise box as high as possible in jaws, remove oil cellar, block across pedestal binder and under journal block on bottom frame rail under ends of arch equalizer, or on top frame rail under spring saddle to relieve weight from box, block on top of the box under frame or on top of ends of arch equalizers under frame at wheel ahead of disabled wheel to throw the weight on these wheels diagonally, and carry engine up level, get off the block with disabled wheel and chain from back corner of engine frame on disabled side, across above draw bar, to opposite forward corner of tank frame, cut out driver brake and proceed. NOTE.—If impossible to carry weight diagonally, use a tie or piece of rail and lash it down to deck of engine and block under end on deck of tank, while the disabled wheel is on wedge, throwing weight on tank.

Q. 250.—What if forward driving wheel or axle broke on Mogul or Consolidation engine?

A.—Take down the necessary side rods and corresponding rods, jack or snub up the axle and handle same as for broken tire except it is not considered necessary to remove oil cellar (merely block between cellar and pedestal, as the journal is ruined anyway) cut out driver brake.

Q. 251.—What if main wheel or axle [illegible]?

A.—All side rods must be taken down. Handle journal same as above, clamp valve centrally on seat, disconnect valve [illegible] take out cylinder cocks and block crosshead ahead, carrying main rod in [illegible]

Q. 252.—What if back wheel or axle [illegible] on Mogul or Consolidation engine?

A.—Remove back section side rods, [illegible] sides, and handle same as for broken [illegible] after you get journal snubbed up as [illegible] as possible. Cut out brake.

Q. 253.—What if axle or wheel broke [illegible] Bissel truck?

journals we must not lose sight of the fact that where the driving journals propel the engine, the engine or car, as the case may be, propels all other journals. The movement of all the journals outside of driving journals is produced by the pressure of the brass against the journal, as shown in Fig. 3, and as this pressure is always in the direction of rotation, it is plain that were grease used in this instance the edge of the brass pressing against the journal would act as a scraper to carry off the lubricant. The question may be asked "Why doesn't the brass act in this manner in the case of oil?" The reason of this has been previously explained, in that the adhesion of oil to a bearing is greater under all circumstances than that of grease. Again, there is no journal that constantly remains in intimate contact with its bearing. On the contrary, as a journal revolves, regardless of the amount of weight upon it, there is always an occasional separation between the journal and its bearing, due to either some inequality in the track or inequality in the wheels, and it is during this interval of separation that the oil can readily be carried between the journal and its bearing.

HOT BEARINGS.

Before leaving the subject of journal lubrication it might be well to mention some of the causes of hot bearings. Of course, if the causes are known, the remedy will be obvious. Principal among causes for hot driving journal bearings where grease is used is the poor fit of the perforated plate previously mentioned. Next will come defect in the brass that will cause a portion of the brass to act as a scraper, cutting away the lubricant. This may be caused either by the brass having become heated at some time and then closed so as to pinch the journal, to a crack in the brass, or to the manner in which brasses are grooved out. In this latter or grooving proposition we are taking issue with the practice advocated by the inventors of the grease cellar, but in this instance we believe our position to be well taken and capable of defense.

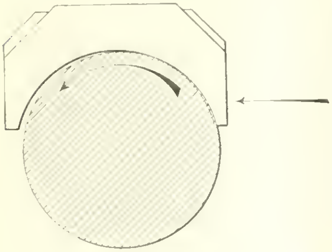


FIG. 3.

Fig. 4 is an illustration of the grooving advocated and practiced in many shops. Of course, in this instance, the primary idea is that grease will be carried up into these grooves and from them distributed to various parts of the journal; also, that these grooves tend to act as receptacles or storage reservoirs for grease, which, when the journal becomes warm, will melt and act as a lubricant. That this theory is entirely incorrect is proven by the fact that there is never any grease found in these grooves when a journal is dropped on account of same running hot. On the contrary, the grooves, instead of acting as receptacles, simply form passages whereby the grease can be carried from one side of the brass to the other without ever coming in intimate contact with the journal at all. The grooves also materially reduce the bearing area, and, last, but not least, every individual groove can, under certain conditions, act as a scraper to scrape away the grease from some one portion of the journal, and thereby set up the condition previously mentioned, which will ultimately result in the heating of the entire journal.

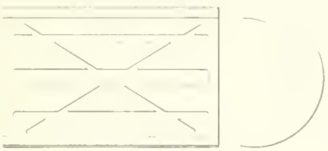


FIG. 4.

As shown in Figures 1 and 2, there is absolutely no need of this grooving or of this reduction in bearing area. In fact, repeated tests have proven that if the entire bearing area of any bearing, but especially a driving box bearing, be left intact, that much better results will be obtained than by having it cut away with numerous grooves. We have often been tempted to ask the advocates of the grooved bearings why the same system was not adopted in crank pin bearings.

CRANK PIN LUBRICATION.

Modern practice commends the use of grease in crank pin lubrication, same as in driving journal lubrication, for practically the same reasons. As we are all aware, the lubricant, grease, is introduced between the crank pin and the brass in practically the same manner as obtains with the driving box, except in this instance the grease is introduced from above instead of below.

In crank pin lubrication the action of the bearing on the journal is practically the same as in driving journal lubrication, in that with the locomotive moving forward when the piston is pulling, the back half of the main rod brass is forced against the crank pin, while the front half is away from it, thereby offering an opening which allows the grease to be carried between the brass and the pin. As the revolution is completed, the pressure is reversed, and the grease is now carried around, completing the cycle.

In order to have perfect crank pin lubrication conditions should be right. Such conditions usually obtain on side rods where the ordinary round bushing is used, and, therefore, little trouble is ever experienced with side rod bearings running hot. Of course, the fact that each side rod bearing only takes its proportion of the load has considerable to do with their running cool. This fact should also be taken into consideration by those whose duty it is to fit up main rod brasses, so that the condition of the main rod brass or bearing may approach as nearly as possible that of the side rod. This means that the brasses should not only be a good fit on the pin, but also a perfect fit in the strap. As the top and bottom of the main rod brasses occupy the same relative positions, so far as the direction of force and lubrication is concerned, as the front and back sides of the driving box brasses, the tops and bottoms of the main rod brasses should be cut away as shown by the dotted lines in Figure 5, so as to afford an opportunity for the grease to get down on the pin and be carried around as the pin rotates. Unless this provision is made, the edges of the brasses will act as scrapers to scrape the grease off just the same as on the driving box.

HOT PINS.

Ordinarily, the cause of any hot pin can be determined by careful investigation. As a rule, however, when pins are reported as running hot the roundhouse man simply takes down the rod reported, and if the brasses all show to be tolerably good fit, he does not look further for the trouble, but probably eases the brasses a little at the top and bottom, and puts the rod up again. As the principal trouble obtains in back ends of main rods, we will confine our remarks entirely to this one pin, although the same remarks will apply to other pins under same conditions.

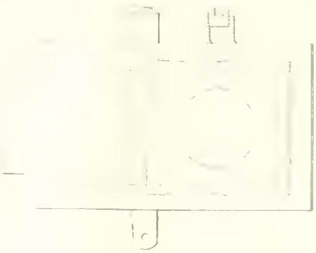


FIG. 5.